

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course : Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTMOE505B - RENEWABLE ENERGY SOURCES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following best explains why solar energy is considered a sustainable energy source?
a) It can be stored easily for long durations
b) It is uniformly available across the globe
c) It does not get depleted with use and is environmentally friendly
d) It requires no technological advancement for effective usage
- 2 The angle between the sun's rays and a line perpendicular to the earth's surface is called:
a) Zenith angle
b) Declination angle
c) Solar altitude angle
d) Hour angle
- 3 Which of the following is not a concentrating solar collector?
a) Parabolic trough collector
b) Heliostat
c) Flat plate collector
d) Paraboloid dish
- 4 The efficiency of a flat plate collector primarily depends on:
a) Mirror quality
b) Fluid color
c) Absorber surface and insulation
d) Shape of the collector
- 5 Which of the following factors significantly affects the efficiency of a photovoltaic cell in real-world conditions?
a) Wire thickness in the inverter
b) Surface reflectivity and cell temperature
c) Panel color and ambient humidity
d) Orientation of water piping system
- 6 Which of the following is not a thermal application of solar energy?
a) Solar water heating
b) Solar space cooling
c) Photovoltaic electricity generation
d) Solar drying
- 7 Which combination of factors is most crucial in determining the feasibility of a wind energy installation at a particular site?
a) Soil type, wind color, and wind noise
b) Average wind speed, wind direction pattern, and terrain roughness
c) Latitude, proximity to ocean, and type of trees nearby
d) Population density, temperature range, and rainfall
- 8 Biomass energy is derived primarily from:
a) Fossil fuels
b) Plant and animal waste
c) Sunlight
d) Wind
- 9 Ocean Thermal Energy Conversion (OTEC) works on the principle of:
a) Salinity gradient
b) Temperature difference between surface and deep water
c) Wave motion
d) Tidal range
- 10 The main disadvantage of geothermal energy is:
a) High emissions
b) Not renewable
c) High capital cost and location limitation
d) Requires sunlight

- 11 Which renewable source has the highest energy density? a) Solar b) Wind c) Biomass d) Nuclear
- 12 Which instrument is used to measure solar radiation? a) Barometer b) Pyranometer c) Anemometer d) Hygrometer

Q2. Solve the following.

- A) Explain the spectral distribution of solar radiation and the factors affecting its attenuation in the Earth's atmosphere. 6
- B) Describe the methods of measuring solar radiation and the properties of opaque and transparent surfaces. 6

Q3. Solve the following.

- A) Compare flat plate solar collectors with concentrating type collectors. 6
- B) Explain different tracking mechanisms used in solar concentrators. 6

Q4. Solve Any Two of the following.

- A) Explain photovoltaic conversion and its applications. [6
- B) Write short note on solar drying and solar still. 6
- C) Describe the use of solar energy in space heating and cooling. 6

Q5. Solve Any Two of the following.

- A) Discuss the types of wind mills and explain any one. 6
- B) Explain site evaluation for bio-conversion and types of biomass resources. 6
- C) Describe the biomass conversion processes. 6

Q6. Solve Any Two of the following.

- A) Write a short note on Ocean Thermal Energy Conversion (OTEC). 6
- B) Explain the working of a geothermal power plant. 6
- C) Describe the scope and limitations of nuclear energy as a renewable source. 6

*** End ***